

Stellarocq: Machine-Checked Certificates for Stellarator Equilibria

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Abstract

VMEC judges its output by its own convergence metric, which measures the retained Fourier harmonics of its forces and not the field reconstructed from its output file. Stellarocq states properties of that field as machine-checked theorems. A Rocq development reconstructs the field from the exact numbers of a VMEC++ output file and proves that the residual it evaluates at a free radius is the field's ideal-MHD force; on the grid's surfaces it evaluates VMEC's discrete form of that force. Bounds, Fourier harmonics, flux-surface integrals, the Mercier criterion and the existence of discrete solutions are decided by checkers extracted from their soundness proofs. On manufactured problems the certified error of the collocated half-grid rule, known to twelve digits, falls about fourfold per doubling of the surfaces. On VMEC++ equilibria the certificates show a resonant force harmonic falling only 1.2 to 1.4-fold per doubling, a free-boundary pressure jump that refinement barely reduces, a Mercier criterion whose sign at coarse resolution depends on the radial differencing, and a W7-X force set by the Fourier truncation. The development also proves that Landreman's explicit three-dimensional fields, counterexamples to Grad's conjecture, are ideal-MHD equilibria; VMEC++ converges to a member of one of his families at about second order and does not converge to a member of the other.

1 Introduction

The Variational Moments Equilibrium Code (VMEC) [1, 2] computes three-dimensional ideal-MHD equilibria with nested flux surfaces by minimizing the plasma energy over Fourier representations of the surfaces. It and its reimplementations VMEC++ [3, 4] sit at the centre of stellarator design and optimization. A run ends with a *wout* file: Fourier coefficients of the surfaces on a radial grid, profiles, and derived quantities such as the Mercier stability terms, all in binary64 floating point.

What such a file establishes about the field it describes is judged by convergence metrics internal to the solver and by diagnostics recomputed in floating point, and those judgements can mislead in ways the file does not show. The normalized force residual a run drives below 10^{-12} concerns the Fourier harmonics of the forces the solver balances, while the force of the field reconstructed from the file, at points of its surfaces, can be set by the angular truncation and not fall as the radial grid is refined. A Mercier term computed from radial differences of the file's profiles can change sign when the differences are taken another way. A rational surface can carry a current sheet that the solution resolves more slowly than anything else as the radial grid is refined. Each of these is a statement about the reconstructed field, and each can be decided exactly.

Stellarocq decides them as theorems. The object of every statement is the field reconstructed from the exact numbers in the file by the solver's own rule between nodes and half points, extended to every radius by cubic Hermite interpolation. A claim about that field, such as a bound on its force at a set of points, is decided by a checking function: a program written in the Rocq proof assistant [5], proven once for all inputs to imply the claim whenever it returns `true`, and extracted to OCaml [6] to run as compiled code. The checking functions compute in interval arithmetic,

whose results contain the exact values however the floating-point rounding falls, with real-number semantics and intervals from Coquelicot [7], Flocq [8] and CoqInterval [9]. A certificate carries the file’s numbers and the claimed bounds, so anyone can check it again with the released checker without trusting the program that wrote it.

Section 2 defines the reconstructed field, proves that the residual Stellarocq evaluates at a free radius is its ideal-MHD force, and gives exactly how its Jacobian and metric differ from the ones VMEC forms. Section 3 describes the checking functions, for bounds at points and over cells, discrete Fourier harmonics, flux-surface integrals, the Mercier criterion and the existence of discrete solutions, and how the development that proves them is built. Section 4 applies them to VMEC++ and to Landreman’s explicit three-dimensional fields [60], counterexamples to Grad’s conjecture both of whose families the development proves to be ideal-MHD equilibria, so that a solver’s distance from them is a true error.

The certificates make such effects proven numbers about the reconstructed field. On manufactured problems the certified error of the collocated half-grid rule, known to twelve digits, falls about fourfold per doubling of the surfaces. On a rational surface the resonant harmonic of the force falls only 1.2 to 1.4-fold per doubling of the surfaces, while the axisymmetric harmonics fall fourfold. At the free boundary the pressure jump falls by factors of only 1.05 to 1.14 per doubling of the surfaces, for every vacuum field between the answers of two independent solvers. On W7-X, converged to $f_{sq} < 10^{-12}$, the certified bound on the force at points of its surfaces does not fall as the surfaces double and falls as the angular resolution rises. At coarse resolution the certified Mercier criterion takes either sign, depending on how the radial derivatives are taken. VMEC++ converges at about second order to Landreman’s sheared equilibrium away from the axis and does not converge to his equilibrium with $\iota = 2$. Section 5 draws the consequences for users of VMEC, and the tests these results added to VMEC++ close Section 4.

2 The certified object

A wout file holds Fourier coefficients on a radial grid, not a field. Stellarocq defines the field by the solver’s own rule between nodes and half points, extends it to every radius by a cubic Hermite interpolation of its own choosing, and computes the field and the force of ideal MHD exactly from the coordinates that result.

Reconstruction. VMEC keeps the surface coefficients of R and Z on the full radial grid $s_j = jh$ and the stream function λ , the rotational transform ι and every field quantity on the half grid $s_{j+1/2}$. Stellarocq’s `Physics.v` writes the rule that connects them. On a half point s_h between nodes a and b , a coefficient of even poloidal mode number is the mean of its two node values and one of odd mode number is averaged with the $\sqrt{s}$ weight VMEC folds into its Jacobian,

$$c_k(s_h) = \frac{1}{2}(c_k(a) + c_k(b)) \quad (m_k \text{ even}), \quad c_k(s_h) = \frac{1}{2}\sqrt{s_h} \left(\frac{c_k(a)}{\sqrt{s_a}} + \frac{c_k(b)}{\sqrt{s_b}} \right) \quad (m_k \text{ odd}), \quad (1)$$

with radial derivatives the matching difference quotients and angular derivatives exact derivatives of the series $R = \sum_k c_k \cos(m_k u - n_k v)$, $Z = \sum_k z_k \sin(m_k u - n_k v)$. The half-grid λ and ι are the file’s own, Φ' is constant, and the pressure is the input’s profile. For statements at an arbitrary radius each coefficient of R and Z is the cubic Hermite interpolant of its half-point values and slopes, which agrees with the half-grid rule at the half points; λ follows the same parity rule linearly between half points, and ι is linear there.

Field and residual. With $\sqrt{g} = R(R_u Z_s - R_s Z_u)$, the field is $B^u = \Phi'(\iota - \lambda_v)/\sqrt{g}$, $B^v = \Phi'(1 + \lambda_u)/\sqrt{g}$, its covariant components are $B_i = g_{iu} B^u + g_{iv} B^v$, and $\mu_0 \sqrt{g} J^s = \partial_u B_v - \partial_v B_u$. The residual of $\mathbf{J} \times \mathbf{B} = \nabla p$ has the components

$$r_u = -\mu_0 \sqrt{g} J^s B^v, \quad r_v = \mu_0 \sqrt{g} J^s B^u, \quad r_s = (\partial_v B_s - \partial_s B_v) B^v - (\partial_s B_u - \partial_u B_s) B^u - \mu_0 \frac{dp}{ds}, \quad (2)$$

written in `Physics.v` as trees of a deep expression language. Read at a free radius, with every radial derivative that of the Hermite reconstruction, they form the continuum residual. Let $X(s, u, v) = (R \cos v, R \sin v, Z)$ be the reconstructed coordinate map, $\mathbf{e}_s$, $\mathbf{e}_u$ and $\mathbf{e}_v$ its partial derivatives, and $\mathbf{B} = B^u \mathbf{e}_u + B^v \mathbf{e}_v$ the reconstructed field as a function of (s, u, v) .

Theorem 1 (continuum_force_coords) *Let the slots of the residual's expressions hold a state's radii, coefficients, flux, transform and pressure derivative p' , and let (s_0, u_0, v_0) be a point where $\sqrt{g} \neq 0$. There let $\nabla s = \mathbf{e}_u \times \mathbf{e}_v / \sqrt{g}$, $\nabla u = \mathbf{e}_v \times \mathbf{e}_s / \sqrt{g}$ and $\nabla v = \mathbf{e}_s \times \mathbf{e}_u / \sqrt{g}$, and let $\partial_s \mathbf{B}$, $\partial_u \mathbf{B}$ and $\partial_v \mathbf{B}$ be the derivatives of $\mathbf{B}$ along the three coordinate lines through the point. Then the three outputs of the continuum residual there are $\mathbf{F} \cdot \mathbf{e}_s$, $\mathbf{F} \cdot \mathbf{e}_u$ and $\mathbf{F} \cdot \mathbf{e}_v$, with*

$$\mathbf{F} = (\nabla s \times \partial_s \mathbf{B} + \nabla u \times \partial_u \mathbf{B} + \nabla v \times \partial_v \mathbf{B}) \times \mathbf{B} - \mu_0 p' \nabla s. \quad (3)$$

In words, the three outputs are the covariant components of $\mu_0(\mathbf{J} \times \mathbf{B} - \nabla p)$ for the reconstructed field, with $\nabla p = p' \nabla s$: the bracket is the curl of $\mathbf{B}$ in the coordinates $x^j = s, u, v$, since the Cartesian derivative of $\mathbf{B}$ is $\sum_j \nabla x^j \otimes \partial_j \mathbf{B}$, and every quantity in the statement is a formula in the reconstruction's derivatives at the point. `continuum_force` gives the same outputs through the Cartesian curl: for any $\mathbf{G} : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ and $P : \mathbf{R}^3 \rightarrow \mathbf{R}$ differentiable at $X(s_0, u_0, v_0)$ and equal to the reconstructed field and pressure along the three coordinate lines through it, they are $\mathbf{F} \cdot \mathbf{e}_i$ with $\mathbf{F} = (\nabla \times \mathbf{G}) \times \mathbf{G} - \mu_0 \nabla P$. `continuum_force_asym_coords` and `continuum_force_asym` are the two statements for a state without stellarator symmetry, and they rest on the identities at one point `Force.force_coords` and `Force.force_law`.

At a node the solver's rule is used instead. The angular outputs are the formulas above at the outer half point, and the radial output is VMEC's combination of the fields at the two half points, with centred differences for radial derivatives; each of those fields is that of the series with VMEC's half-grid values and slopes (`node_residual`, `node_residual_asym`).

VMEC's products. Every half-point quantity that is linear in the coefficients, R , Z and their derivatives, agrees with VMEC's term for term. The Jacobian and the metric elements are products, which VMEC forms from averages over the two nodes of products of node values, where the reconstruction multiplies the interpolated series. `VmecKernel.v` transcribes VMEC++'s kernels and proves the differences exactly. With r , z the even parts and $\hat{r}$, $\hat{z}$ the odd parts divided by $\sqrt{s}$ of the node values, and Δ the outer node's value less the inner one's, the reconstruction's factor $R_u Z_s - R_s Z_u$ and VMEC's τ satisfy

$$(R_u Z_s - R_s Z_u) - \tau = -\frac{1}{8}(\Delta \hat{r}_u \Delta \hat{z} - \Delta \hat{z}_u \Delta \hat{r}) - \frac{1}{8\sqrt{s_h}}(\Delta r_u \Delta \hat{z} - \Delta z_u \Delta \hat{r}) \quad (4)$$

(`vmec_jacobian_difference`), and each of g_{uu} , g_{uv} and g_{vv} differs from VMEC's by a sum of such products (`vmec_metric_difference`). Where every node difference in the identity is at most $K|h|$, the two Jacobian factors agree to within $\frac{1}{4}(Kh)^2(1 + s_h^{-1/2})$ (`vmec_jacobian_second_order`). The certified field is the reconstruction's, and VMEC's discrete field differs from it by these terms, of second order in the radial step away from the axis. Evaluated in floating point

(`gen/jacobian_gap.py`) on the stellarator-symmetric files of Section 4, the two Jacobians, and so the two $\sqrt{g}$ (`vmec_sqrtg_difference`), differ by at most a relative 8.9×10^{-5} on the half-grid surfaces whose nodes lie off the axis, the largest on W7-X at 49 surfaces near the axis. On the surfaces with $s \geq 1/4$ the difference falls as the surfaces double, on W7-X from 2.3×10^{-5} at 49 surfaces to 1.5×10^{-6} at 193 and on Landreman’s sheared equilibrium from 6.3×10^{-7} at 13 to 1.2×10^{-9} at 200.

Identities. The representation builds some properties in, and for the reconstruction they are theorems: it is solenoidal (`divergence_free`), the $(0, 0)$ coefficient of λ drives nothing (`lambda_gauge`), which exhibits the poloidal relabelling as an exact kernel direction, and the residual has no component along the field (`residual_along_field`). The Euler–Lagrange expressions of VMEC’s energy $W = \int (B^2/2 - \mu_0 p) \sqrt{g}$ at fixed λ, ι , flux and pressure are $-\sqrt{g}$ times the cylindrical components of the force (`energy_gradient_is_force`), so that the forces the solver drives to zero are the ideal-MHD force. `Hypotheses.v` states the physical premises some theorems carry, such as nested flux surfaces and a scalar pressure.

3 Certificates and their soundness

A certificate is a text file that a generator in `gen/` writes from a wout file. It carries every number of the file it uses as an integer mantissa and a power-of-two exponent, so no rounding happens between the file and the certificate, together with the bounds it claims. The extracted checker recomputes every enclosure from those numbers, and no bound the generator computes is trusted. A check passes when its enclosures, intervals that contain the exact values, lie inside the claimed bounds, and its soundness theorem turns that comparison into a theorem about the field.

Expressions and intervals. `Expr.v` gives an expression two meanings: a value over extended reals, in which division by zero and the square root of a negative number give `Xnan`, and a value over CoqInterval’s floating-point intervals. `ieval_correct` states that the interval contains the extended-real value, so a bounded output interval proves both that the value is a real number and that it lies in the bounds. Because the embedding is deep, one tree carries both meanings. `Deriv.v` differentiates it symbolically with a chain rule proven in the extended-real sense, and subexpressions that the residual’s components share are bound once to environment slots.

Points and cells. A point certificate bounds the residual at points (`check_cert_correct`). A cell certificate bounds a component over a rectangle of two varied slots, two angles or an angle and the radius, by its value at the centre plus enclosures of its derivatives over the rectangle times the half-widths (`check_ccert_correct`); reversed, the same data bounds a component away from zero. The bound tightens as the cells shrink, since each derivative enclosure is taken over its own cell. On one surface of the Solov’ev equilibrium, where the residual reaches 4.1×10^{-7} in floating point, the bound falls from 2.2×10^3 with 8 cells to 7.7×10^{-6} with 8192, about sixteen-fold per fourfold refinement from 128 cells on. `Cover2.v` decides, column by column, that the cells of a verdict cover the rectangle they span (`cells_cover_rect`) or every line they lie on (`cells_cover_lines`).

Harmonics and integrals. A spectral solver balances forces mode by mode: VMEC drives to zero the harmonics, at the modes it retains, of $\sqrt{g}$ times the cylindrical components of the force (`energy_gradient_is_force`). `Project.v` encloses the discrete Fourier harmonic of a covariant component of the residual at a retained mode (`harm_encloses`). That component contracts the

force with a coordinate tangent vector and carries no $\sqrt{g}$ weight, so its harmonic at a retained mode also carries the modes VMEC drops and need not vanish when VMEC's do. `Harmonic.v` integrates exactly the components whose trigonometric degree lies below an equispaced grid's point counts. `Integrate.v` encloses other flux-surface integrals by the midpoint rule over a lattice of cells that the checker lays out itself, with the error read off its own enclosures of the second derivatives over each cell (`integ2_correct`, `integ1_correct`). `MercierRun.v` assembles DShear, DCurr, DWell, DGeod and DMerc as VMEC's `mercier.f90` defines them [10, 11] from integrals over the whole angular torus that the same run computes. Its lattice reaches past each period of 2π , which ends inside the last cells, and the two strips it adds there are bounded through the integrand's enclosures over those cells (`integ2_torus_correct`), so a verdict is a theorem about the criterion of the reconstruction (`merc_run_correct`).

Boxes of states. In `Box.v` every input slot carries a half-width, so one evaluation covers every field whose coefficients lie in a box. The theorem behind the resonant harmonic of Section 4 shows the form every soundness theorem takes:

```
Theorem dharm_correct :
  forall c, check_dharm c = true ->
  forall xs, in_box (hc_ms c) (hc_ds c) xs ->
  contains (I.convert (harm_total c))
    (Xreal (dsum2 (hc_Nu c) (hc_Nv c) 0 0 (harm_fun c xs))).
```

For a certificate `c` whose check passes and every state `xs` in its box, the equispaced sum of the component times the harmonic's kernel over the certificate's grid of angles is a real number, and it lies in the interval `harm_total c` that the same computation returns.

Discrete equilibria. `Newton.v` proves that a passing interval Newton test in Krawczyk form [12, 13] on any system of expressions gives exactly one zero in its box: $x - AF(x)$ maps the box into itself as a contraction, and an approximate inverse A makes its fixed point a zero of F . `Colloc.v` collocates the residual over a band of surfaces as such a system, with the surface coefficients as unknowns (`colloc_correct`). `Wide.v` evaluates the system's outputs at the centre of the box in floating point of any precision.

Proof engineering. The theorems and checkers of this paper rest on 39 files and about 27 000 lines of Rocq with about 800 theorems and lemmas, part of a development of 167 files that **make all** builds, with the checker, in about four minutes with six parallel jobs. Each kind of claim is decided by a Rocq function on syntax, proven sound once for all inputs, and the kernel never evaluates a certificate, so no proof term grows with one and a check costs what compiled OCaml costs. The physics enters only through expression trees, which `Deriv.v` differentiates and every checker evaluates in intervals proven to contain their one extended-real meaning: binary64 intervals on Rocq's primitive floats (`ieval_correct`) and, for the checks at 120 to 256 bits, CoqInterval's floats with integer mantissas of any precision (`weval_correct`). A theorem about what a tree means thus serves every certificate that evaluates it.

Trust base. The kernel checks the proofs; what runs is the OCaml extraction of the checking functions, linked with a driver that parses certificates into the extracted data types, divides the checking among processes and prints verdicts. The trusted components are the Rocq kernel and extraction, which realizes Rocq's natural numbers as OCaml's native integers (`ExtrOcamlNatInt`),

Check	Size	Time
points, free-boundary jump	NESTOR’s grid	0.1–3 s
points, $\mathbf{B} - \mathbf{B}^*$ of Landreman’s field	64×32 points	3–18 s
cells, Solov’ev residual on a surface	8–8192 cells	up to 15 s
cells, free-boundary jump over the boundary	512×128 cells per period	72 and 78 min
harmonic over a box of states, binary64	64×32 points	2 s
harmonic, rows at 256 bits	64×32 points	1.4–2.1 h
Mercier criterion, tokamak	four coverings of 2048 cells	2–3 min
Newton existence, 120 or 128 bits	up to 423 unknowns	up to 46 min
W7-X points and projection	4096 points, 288–800 modes	21 min–3.4 h

Table 1: Time to check each kind of certificate.

the OCaml toolchain, the implementation of Rocq’s primitive 63-bit integers and binary64 floats, the driver’s 3 900 lines of OCaml with its parser and its division of the work, and the correspondence between a certificate’s numbers and the file or formulas they are taken from. `gen/verify_cert.py` checks that correspondence slot by slot against the wout file for the residual, cell, point, surface and harmonic certificates. The slots no wout holds, the vacuum pressure of the free-boundary certificates and the parameters of Landreman’s member, stay in the trust base with the data of the Newton certificates. The classical constructions behind Rocq’s real numbers appear only in specifications, and extraction directives replace them by values that fail when consulted. The theorems named here rest on the two axioms of the standard library’s classical Dedekind reals, excluded middle, functional extensionality and the specifications of the primitive operations. The build pins Rocq 9.1.1 with its standard library 9.2.0, CoqInterval 4.11.4, Flocq 4.2.2, Coquelicot 3.4.5, MathComp 2.4.0 and OCaml 4.14.2.

Cost. Checking takes seconds to hours (Table 1). A point certificate evaluates the residual once per point, a cell certificate its derivatives over each cell and a harmonic every point of its grid, so the cost grows with the number of modes, the number of points or cells, and the precision. A harmonic whose rows are evaluated at 256 bits in software floating point takes hours where the same harmonic in binary64 takes seconds. The Mercier criterion integrates over the whole torus, and its cells must be narrow against the highest mode there for the enclosures over them to stay bounded. Four coverings of 2048 cells take two to three minutes for the tokamak `up_down_asym`, while for W7-X, whose toroidal numbers reach 60 on the torus, a covering of 64×16 cells leaves enclosures unbounded. The checker divides a certificate’s points, cells or rows among processes.

4 Results

Table 2 collects the certified results with the theorem each rests on. The manufactured problems are Stellarocq’s collocated systems for the manufactured mappings of a VMEC++ pull request, still open, that adds a force source to the solver [29]; every other case is a VMEC++ run or a file from its test data. The runs of Landreman’s equilibrium, of the rippled tokamak and of W7-X are VMEC++ 0.4.7, and the free-boundary and Mercier runs use the branch of [28], which reads NESTOR’s vacuum field. A number called certified is one that a passing verdict proves. On the surfaces of a run the residual is the node rule of Section 2, VMEC’s discrete form of the force, which is consistent with the continuum residual to second order away from the axis (`node_consistent`).

Case and surfaces	Certified quantity and unit	Values, surface by surface	Theorem
manufactured, 3D; 9, 17, 33, 65	$\max x_h - x^* , 10^{-7}$	27.34, 7.169, 1.830, 0.4607	<code>colloc_correct</code>
asymmetric; 9, 17, 33	the same, 10^{-6}	26.91, 7.207, 1.863	<code>colloc_correct</code>
Solov'ev; 1, 2, 3	zero of the collocated residual	one in each box, radius 6.0×10^{-10} to 7.7×10^{-9}	<code>colloc_correct</code>
rippled tokamak; 65, 129, 257, 513, 1025	(2, 1) harmonic of r_s at $\iota = 1/2$, 256 bits, 10^{-5} T^2	8.478, 8.762, 7.003, 5.197, 3.759	<code>wdharm_b_correct</code>
the same; 65, 129, 257	the same over a box of relative width 10^{-14} , 10^{-5} T^2	[8.37, 8.59], [8.34, 9.18], [5.35, 8.66]	<code>dharm_correct</code>
CTH-like; 25, 49, 97, 193, 385	largest jump of $\mu_0 p + B^2/2$ on NESTOR's grid, 10^{-4} T^2	7.56, 7.20, 6.44, 5.64, 4.97	<code>check_bpcert_correct</code>
the same	bound with the vacuum-pressure sweep five times wider, 10^{-3} T^2	1.87, 1.89, 1.81, 1.73, 1.66	<code>check_bpcert_correct</code>
the same; 25, 385	bound over the whole boundary, 10^{-3} T^2	8.47, 10.51	<code>check_btcert_correct</code>
<code>up_down_asym</code> ; 17	DMerc at $s = 0.125$, reconstruction and file's differences, 10^{-3}	$[-1.371, -1.205]$, $[1.711, 1.875]$	<code>merc_run_correct</code> , <code>merc_run_q_correct</code>
Landreman, sheared; 13, 25, 50, 100, 200	bound on the largest component of $\mathbf{B} - \mathbf{B}^*$, middle and outermost surfaces, 10^{-7} T	132, 29.3, 5.39, 1.12, 0.253; 82.5, 20.0, 5.67, 1.02, 0.212	<code>check_bpcert_correct</code>
the same	bound on the spread of ψ^* about a level set, 10^{-9}	232, 49.5, 17.7, 1.70, 0.171	<code>check_bpcert_correct</code>
Landreman, $\iota = 2$; 13, 25, 50, 100	floor on a component of $\mathbf{B} - \mathbf{B}^*$ at one point, 10^{-4} T	7.97, 2.04, 17.3, 1.63	<code>check_bpcert_correct</code>
W7-X; 49, 97, 193	bound on the largest component of the node residual, 10^{-2} T^2 , and on its largest retained harmonic, 10^{-4} T^2	7.71, 7.73, 7.82; 5.49, 5.49, 5.46	<code>check_cert_correct</code> , <code>harm_encloses</code>

Table 2: Certified results and the soundness theorem behind each. Intervals are enclosures rounded outward, bounds are rounded up and floors down, and a single value is certified to within one unit of its last digit.

The order of the half-grid rule. On manufactured problems away from the axis the certified error of the half-grid rule falls about fourfold per doubling of the surfaces and is known to twelve digits. `HalfGrid.v` proves the node rule second-order consistent with the residual of the continuously reconstructed field (`node_consistent`); near the axis, where the innermost interval is read linearly, the rule is first order in the radial derivatives. Manufactured solutions [15] measure the order. The mappings are the $m \leq 1$ ones of the manufactured-solution test of [29], with three field periods. The three-dimensional one is $R = 1 + p_0 s + p_1 s^2 - 0.005 \cos 3\phi + \sqrt{s} (0.05 \cos u + 0.005 \cos(u - 3\phi))$ and $Z = 0.005 \sin 3\phi + \sqrt{s} (0.05 \sin u - 0.005 \sin(u - 3\phi))$ with $p_0 = -3.32 \times 10^{-4}$ and $p_1 = -8.06 \times 10^{-6}$, $\iota = 0.42 + 0.13s$, $p = 1.6 \times 10^5 (1 - s) \text{ Pa}$ and a stream function with harmonics up to $m = 2$, and the one without stellarator symmetry adds $0.004 \sin 3\phi + \sqrt{s} (0.006 \sin u + 0.002 \sin u \cos 3\phi)$ to R and $0.003 \cos 3\phi + \sqrt{s} (0.006 \cos u + 0.002 \cos u \cos 3\phi)$ to Z . The file that defines them, `examples/manufactured_solution.py` of that pull request at commit `fbd696`, is copied into Stellarocq's `gen/`, and the generator reads every constant of the problem from it.

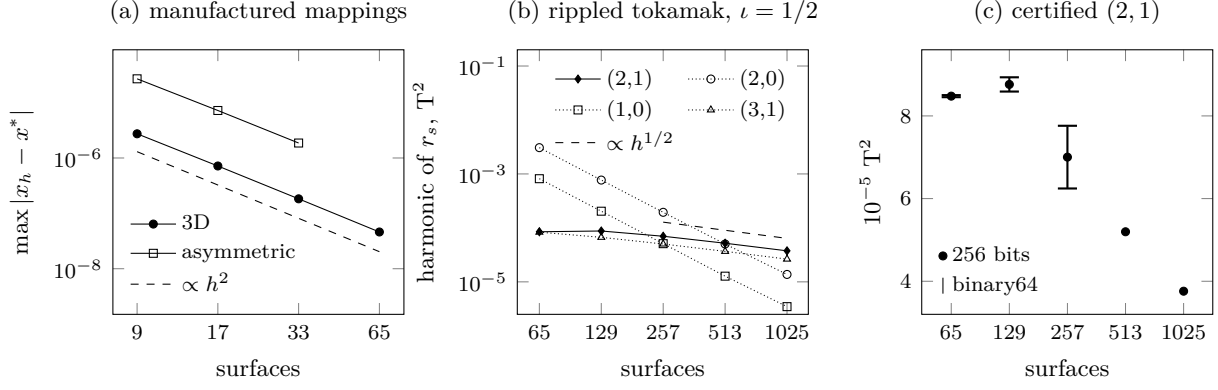


Figure 1: (a) The certified error of the half-grid scheme on the manufactured mappings; the dashed line falls as h^2 . (b) Harmonics of the radial force r_s on the surface $\iota = 1/2$ of the rippled tokamak: the resonant (2,1) harmonic, certified at 256 bits (solid), and the floating-point (2,0), (1,0) and (3,1) harmonics of the same reconstruction (dotted); the dashed line falls as $h^{1/2}$. (c) The (2,1) harmonic on a linear scale: its binary64 enclosures at the files' states at 65, 129 and 257 surfaces, and its 256-bit enclosures, narrower than the dots, at 65 to 1025.

A manufactured mapping on the annulus $1/4 < s < 1$ is not an equilibrium. Its continuum residual at the collocation points is a source f , and the discrete solution is the zero x_h of $r - f$ over the interior nodes, with λ given so that the problem sits on the quotient by the poloidal gauge. Every datum of the problem, the coefficients it holds fixed, λ , the transform and the source, is the mapping's own value carried to 120 bits. On systems of up to 423 unknowns the interval Newton test puts exactly one zero in a box a few units of a 58-bit mantissa wide in each unknown, around a centre refined on outputs read at 128 bits, so the error $\max |x_h - x^*|$ is enclosed to within 5×10^{-20} , a relative 10^{-12} (Table 2). It falls by 3.81, 3.92 and 3.97 per doubling for the three-dimensional mapping and by 3.73 and 3.87 for the one without stellarator symmetry (Figure 1a). The same problem with its data rounded to binary64 has an error that differs in the eighth digit.

The collocated system is the half-grid residual of the reconstruction at collocation points. VMEC's own discrete equations come from the variation of the energy with spectral condensation and a constraint on the $m = 1$ modes, and they form the Jacobian and metric from the products whose difference from the reconstruction's `VmecKernel.v` gives exactly, at second order in h . The annulus keeps the axis, where the rule is first order, out of the problem; the first-order radial convergence that [16] report for VMEC is that of the force error of VMEC's own solutions averaged over $0.1 \leq s \leq 0.99$ of a discrete problem that contains the axis. On an equilibrium the collocated system has solutions as well: one, two and three surfaces of the Solov'ev equilibrium [14] each have exactly one zero of the collocated residual in a box of radius 6.0×10^{-10} to 7.7×10^{-9} around a centre that Newton's method moved 1.8×10^{-6} to 3.0×10^{-6} from the file's coefficients, and five surfaces do not pass the test.

A forced current sheet. At a rational surface the discrete solution converges more slowly than anywhere else: the resonant harmonic of the force falls by only 1.2 to 1.4 per doubling of the surfaces. On VMEC++'s circular tokamak ($R_0 = 6$, $a = 2$) with $\iota = 0.9 - 0.64s$, which is $1/2$ at $s = 0.625$, and a (1,1) boundary ripple of amplitude 0.03, toroidal coupling drives a (2,1) harmonic of the radial force on the rational surface. Its certified value is the rectangle rule for the integral of $r_s \cos(2u - v)$ on a 64×32 grid of angles, in T^2 like r_s , against $1.2 \times 10^3 T^2$ for the integral of B^2

there. With each row of the grid evaluated and summed at 256 bits, `wdharm_b_correct` encloses it at the states VMEC++ reaches with a tolerance of 10^{-18} at 65 to 1025 surfaces, to within a relative 2.1×10^{-12} (Table 2, Figure 1c), and it is linear in the ripple. After a rise of 3.3 per cent from 65 to 129 surfaces it falls by 1.25, 1.35 and 1.38 per doubling, approaching the factor $\sqrt{2}$ of a decay as $h^{1/2}$. A decade looser, at 10^{-17} , the harmonic differs in floating point by at most 1.3 per cent, and at 10^{-15} by up to 6.5 per cent, while its fall per doubling stays between 1.20 and 1.41 at every tolerance. The axisymmetric harmonics (2, 0) and (1, 0) of the same reconstruction fall 3.7 to 4.0-fold per doubling in floating point, and the (3, 1) sideband 1.25 to 1.39-fold (Figure 1b). The same ripple on a transform profile with no low-order rational surface nearby leaves nothing behind.

Across the surfaces the harmonic forms a layer. In floating point it peaks within a radial step of the rational surface and changes sign outside it, and the distance from its peak to its trough shrinks about as $h^{1/2}$, from 0.117 to 0.040 in s between 129 and 1025 surfaces, while its zero closes on the rational surface from 0.054 to 0.024. This is the response of a nested-surface field at a rational surface [17, 18, 19], whose ideal limit is a current sheet and which [20] studied in VMEC; three-dimensional equilibria with current sheets and stepped pressure exist [21, 22]. SPEC [23] makes the sheet explicit. With two Beltrami volumes separated by an ideal interface at the rational surface it is a jump of 1.47×10^{-5} in the covariant poloidal field at ripple 0.03, as in the equilibria of [24], and with the rational surface inside a relaxed volume SPEC’s field-line tracing finds a (2, 1) island whose width grows from 0.107 to 0.173 in the volume’s radial coordinate as the ripple goes from 0.0075 to 0.03, as the ripple to the power 0.35.

In binary64 interval arithmetic (`dharm_correct`) the harmonic lies between 5.3×10^{-5} and 9.2×10^{-5} for every state within a relative 10^{-14} of the files’ coefficients at 65, 129 and 257 surfaces. These enclosures widen about fourfold per doubling, from 2.5 to 47 per cent of the value, as second radial differences amplify a perturbation by h^{-2} , and beyond 257 surfaces a box of that width no longer separates the harmonic from zero. On the rational surface the collocated Newton test, its centre refined on outputs read at 128 bits, certifies a discrete solution at 33 and at 65 surfaces, where the Jacobian’s condition number is 1.4×10^6 and 3.0×10^6 in floating point.

Free-boundary balance. At the free boundary the balance VMEC++ imposes leaves a pressure jump that radial refinement barely reduces. At a free-boundary equilibrium $\mu_0 p + B^2/2$ is continuous across the plasma boundary, where outside it is the pressure of the full vacuum field. VMEC++ computes that field with NESTOR [25, 2], and BIEST’s virtual-casing field [26, 27] gives it independently. The certificate sweeps the vacuum pressure over the segment between the two answers, $P_{\text{vac}} = P_{\text{mid}} + t \delta P$ with $t \in [-1, 1]$, and takes the plasma side as VMEC’s extrapolation of the two outermost half points to $s = 1$. The sweep variable is a slot of the box, which `check_bpcert_correct` and `check_btcert_correct` quantify over separately at each point and in each cell, so every statement below holds for each vacuum pressure whose value at each point lies between the two answers there.

On the CTH-like test case, where $B^2/2$ averages 0.18 T^2 on the boundary, the largest jump on NESTOR’s grid is enclosed in $[7.556, 7.561] \times 10^{-4} \text{ T}^2$ at 25 surfaces and in $[4.9647, 4.9673] \times 10^{-4} \text{ T}^2$ at 385 (Table 2, Figure 2a). Each enclosure lies below the one before it, so the jump does fall, but only by 1.05 to 1.14 per doubling of the surfaces. The two answers differ by up to $6.1 \times 10^{-4} \text{ T}^2$ on the grid. With the sweep widened to five times that difference the largest jump is at most $1.89 \times 10^{-3} \text{ T}^2$, a hundredth of $B^2/2$, at every resolution. Covered over the whole boundary by 512×128 cells per field period, with the vacuum side the trigonometric interpolant of each answer’s grid, the jump is at most $8.47 \times 10^{-3} \text{ T}^2$ at 25 surfaces and 1.051×10^{-2} at 385. Between grid points that interpolant sets the bound, and radial refinement leaves its angular resolution as it is.

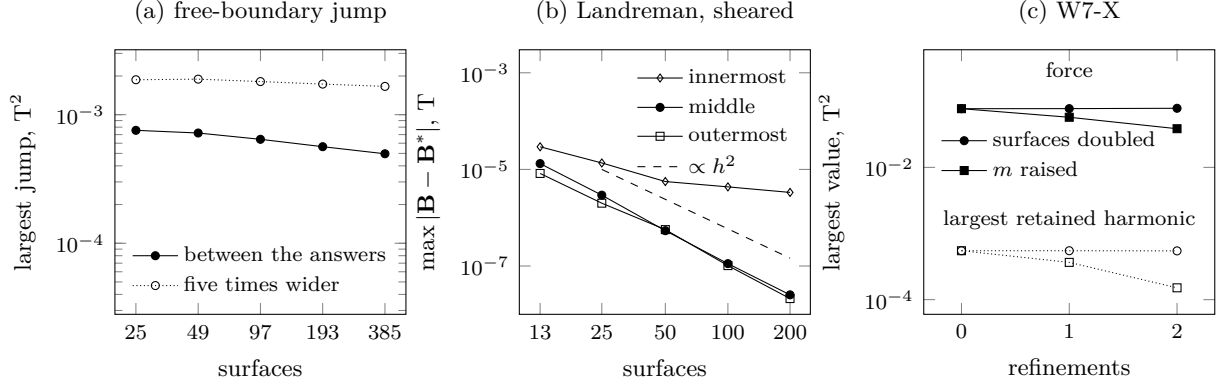


Figure 2: (a) The largest jump of $\mu_0 p + B^2/2$ against the vacuum pressure on NESTOR’s grid of the CTH-like case: its certified value for every vacuum pressure between NESTOR’s and BIEST’s answers, whose enclosures are narrower than the dots, and its certified bound for every vacuum pressure within five times their disagreement. (b) The largest component of $\mathbf{B} - \mathbf{B}^*$ against Landreman’s sheared member on the innermost, middle and outermost half-grid surfaces, certified bounds; the dashed line falls as h^2 . (c) W7-X: certified bounds on the largest component of the node residual $\mu_0(\mathbf{J} \times \mathbf{B} - \nabla p)$ on the surfaces $s = 1/4, 1/2$ and $3/4$ (filled) and on its largest harmonic at a retained mode (open), as the surfaces double from 49 to 193 at $m \leq 11$ and as the angular resolution rises from $m \leq 11$ to $m \leq 19$ at 49 surfaces.

The jump lies almost wholly beyond the Fourier harmonics through which VMEC imposes the balance. In floating point at 49 surfaces its part in the retained harmonics is at most $2.4 \times 10^{-5} T^2$ and the rest 7.2×10^{-4} ; the retained part falls 3.3 and 1.9-fold per doubling from 25 to 97 surfaces, while with NESTOR’s poloidal grid fixed, retaining 5 to 12 poloidal harmonics moves the jump between 4.0×10^{-4} and 1.5×10^{-3} without reducing it.

The Mercier criterion at coarse resolution. At coarse radial resolution the sign of the Mercier criterion depends on how its radial derivatives are taken, and the certificates prove both signs. At $s = 0.125$ of `up_down_asym` on 17 surfaces, with 2048 poloidal cells, the criterion of the Hermite reconstruction is certified unstable: `DMerc` lies in $[-1.371, -1.205] \times 10^{-3}$ and the current term in $[-3.724, -3.716] \times 10^{-3}$, where the file’s own difference quotients give a stable $+1.8 \times 10^{-3}$ and a current term of -6.1×10^{-4} . `MercierFile.v` takes the four flux-function derivatives from the file’s centred differences of its half-grid profiles instead, each an exact dyadic, and integrates the same angular integrands (`merc_run_q_correct`): `DMerc` then lies in $[1.711, 1.875] \times 10^{-3}$, which contains the file’s value. The two certified answers differ only in the radial derivatives, so at 17 surfaces those set the sign. From 33 to 257 surfaces both current terms tend to zero with the file’s sign, the certified one from $[-1.092, -1.083] \times 10^{-3}$ to $[-3.7, -1.8] \times 10^{-5}$, and the certified `DMerc` is positive at each. The equilibrium carries no pressure, so in the continuum its current and geodesic terms vanish and `DMerc` reduces to the shear term, which is never negative: the instability certified at 17 surfaces belongs to the discretization.

An exact equilibrium. Landreman’s families [60] are three-dimensional equilibria with nested flux surfaces and smooth field and pressure, which he presents as counterexamples to Grad’s conjecture [17]. Their field $\mathbf{B}^*$ and flux label ψ^* are elementary functions of the Cartesian position, so a solver’s distance from them is a true error. `Landreman.v` and `LandremanSheared.v` prove that

both consist of ideal-MHD equilibria: the family with $\iota = 2$ for all positive semiaxes wherever the radicand of $\mathbf{B}^*$ is positive, and the sheared family for all positive ϵ and λ and every S wherever (x, y) lies off the segment $x = 0, |y| \leq \sqrt{\epsilon}$. In each the field is solenoidal (`iota2_divergence`, `sheared_divergence`), satisfies $(\nabla \times \mathbf{B}) \times \mathbf{B} = \nabla p$ with $p = p_a - 2\psi$ and $p = p_a - \psi/\lambda^2$ respectively (`iota2_force`, `sheared_force`), and is tangent to the level sets of p (`iota2_tangent`, `sheared_tangent`). Two outputs of the checker compare the reconstruction with a member of either family: the cylindrical components of $\mathbf{B} - \mathbf{B}^*(X)$ on a half-grid surface, with $\mathbf{B}$ the reconstructed field and X its position, and $\psi^*(X)$ on a node surface. `LandremanExact.v` proves that the expressions behind those outputs evaluate to the field and flux label of `Landreman.v` and of `LandremanSheared.v` on those domains (`exact_b_iota2`, `exact_b_sheared`).

Against the sheared family VMEC++'s field converges at about second order away from the axis. Its solution of the member with $\epsilon = 0.6$, $S = 2.2$, $\lambda = 2.97$ and $k_b = 0.2$, whose transform runs from 4.34 to 4.37, at 13 to 200 surfaces with $m \leq 11$ and $|n| \leq 10$, is certified at every point of a 64×32 grid of a field period on its innermost, middle and outermost half-grid surfaces, where $|\mathbf{B}^*|$ lies between 0.30 and 0.38 T (Table 2, Figure 2b). The radial grids are not nested, and the middle surface stays near $s = 1/2$, at 0.54, 0.52 and then 0.50. There the certified bound on the largest component of $\mathbf{B} - \mathbf{B}^*$ falls from 1.32×10^{-5} T at 13 surfaces to 2.53×10^{-8} T at 200, by 4.4 to 5.4 per doubling. On the outermost surface it falls from 8.25×10^{-6} to 2.12×10^{-8} T while that surface moves from $s = 0.958$ to 0.997, toward the fixed boundary, whose coefficients are the member's own. The innermost surface moves toward the axis, from $s = 0.125$ to 0.008, and there the bound falls only from 2.94×10^{-5} to 3.34×10^{-6} T. At the points of the same grid the outermost node surface, moving from $s = 0.917$ to 0.995, lies within 2.32×10^{-7} of a level set of ψ^* at 13 surfaces and within 1.71×10^{-10} at 200, where ψ^* is 0.02 on the boundary.

Against the member with $\iota = 2$, whose every surface is rational, VMEC++'s field does not converge. Its runs with $m \leq 9$ and $|n| \leq 8$ at 13, 25, 50 and 100 surfaces end at their limit of 20000 iterations, their force residual wandering between 7×10^{-14} and 4×10^{-10} instead of approaching the tolerance of 10^{-18} . At one point of the middle half-grid surface a component of $\mathbf{B} - \mathbf{B}^*$ is at least 7.97×10^{-4} , 2.04×10^{-4} , 1.73×10^{-3} and 1.63×10^{-4} T at the four resolutions, each a floor that `check_bpcert_correct` proves (Table 2). In floating point 77 to 86 per cent of the difference's spectral power lies in the resonant harmonics. Those bend no field line when $\iota = 2$ on every surface, so the field-line bending that stiffens VMEC's energy against every other harmonic does not act on them.

An optimized stellarator. On W7-X the certified bound on the force is set by the Fourier truncation and not by the radial grid. VMEC++'s solution of W7-X from its test data has a volume-averaged β of 4.0 per cent, $m \leq 11$ and toroidal numbers up to 12 per field period, and converged to $f_{sqr} < 10^{-12}$ at 49, 97 and 193 surfaces. It is certified at every point of a 64×64 grid of a field period on the surfaces $s = 1/4$, $1/2$ and $3/4$ (`check_cert_correct`), and the same verdicts enclose the discrete harmonics of each component there at the 288 modes VMEC++ retains (`harm_encloses`). At every resolution the certified bound on the largest component of the node residual, $\mu_0(\mathbf{J} \times \mathbf{B} - \nabla p)$ by VMEC's radial differences, is close to 7.7×10^{-2} T², and that on its largest harmonic at a retained mode close to 5.5×10^{-4} T², at $m = 10$, next to the edge of the retained band (Table 2). Raising the angular resolution does what doubling the surfaces does not: at 49 surfaces with $m \leq 15$ and $m \leq 19$, toroidal numbers up to 16 and 20, the largest component is at most 5.72 and 3.84×10^{-2} T² and the largest retained harmonic at most 3.66 and 1.51×10^{-4} T² (Figure 2c).

Use in VMEC++. The certified values are regression tests on a branch of VMEC++ [28], whose tolerances are the enclosures: the manufactured-solution errors, the resonant harmonic with the fall of the axisymmetric harmonics, and the free-boundary jump, which reads NESTOR’s boundary vacuum field through an option proposed upstream [30]. A finite difference of VMEC++’s energy equals its unpreconditioned forces contracted with the perturbation, the discrete form of `energy_gradient_is_force`; VMEC++ now tests this [31], and that the energy Hessian is symmetric and DGeod never positive [32]. Its continuous integration now runs VMEC++ on members of both of Landreman’s families on every change and measures, in floating point, its flux surfaces, magnetic axis, field and enclosed current against the exact ones [33]. Differential tests of VMEC++ for invariances the reconstruction satisfies exactly, three of which VMEC++ now runs [34], also exposed an orientation test that the poloidal rotation VMEC applies to an asymmetric boundary can defeat. The test multiplies two $m = 1$ coefficients whose product the rotation can drive to zero, where the invariant is the $m = 1$ determinant $r^{cc}z^{sc} - r^{sc}z^{cc}$. Relabelling the `up_down_asym_current` boundary by $\theta \rightarrow \pi - \theta$ sends educational_VMEC to its iteration cap at $f_{sqr} = 2.07 \times 10^{-2}$ instead of converging to 5.07×10^{-12} . It was reported as vmecpp#788 [35], after which VMEC++ tests the determinant [36], and as educational_VMEC#27 [37].

5 Consequences for users of VMEC

A converged run does not certify its field. VMEC’s normalized force residual measures the harmonics, at the modes it retains, of the forces the solver balances, not the force of the field reconstructed from its file. For W7-X converged to $f_{sqr} < 10^{-12}$, the certified bounds on that force at points of its surfaces and on its largest retained harmonic do not fall as the surfaces double, while both fall as the poloidal and toroidal resolution rises. A convergence study of a stellarator equilibrium therefore has to vary the angular resolution as well as the number of surfaces, and a quantity computed from the field at points inherits an error that the angular truncation sets and the residual does not report. The point and harmonic checks measure that force on any wout file, in seconds at points and in hours for a harmonic at 256 bits (Table 1).

Radial convergence away from rational surfaces says little about their neighbourhood. Near a low-order rational surface the discrete solution forms a current layer whose resonant force harmonic falls only 1.2 to 1.4-fold per doubling of the surfaces, so an equilibrium with such a surface in the plasma converges there far more slowly than elsewhere. Where every surface is rational, as in Landreman’s family with $\iota = 2$, the solution does not converge at all. At coarse radial resolution a quantity built from radial derivatives of the profiles can change sign with the way the derivatives are taken: at 17 surfaces the Mercier criterion of `up_down_asym` is certified unstable for the reconstruction and stable for the file’s own differences, though without pressure it is stable in the continuum, and from 33 surfaces on the certified criterion is stable. A Mercier sign near marginal stability from a coarse run should be confirmed at more surfaces; on a tokamak the certified criterion takes minutes.

At a free boundary the pressure balance barely improves with radial resolution. The jump of $\mu_0 p + B^2/2$ against the vacuum pressure on the CTH-like boundary falls only from 7.6 to $5.0 \times 10^{-4} \text{ T}^2$, a few thousandths of $B^2/2$, as the surfaces go from 25 to 385, and retaining more poloidal harmonics on NESTOR’s fixed grid moves it without reducing it.

6 Related work

Rigorous numerics proves statements about dynamical systems and partial differential equations with interval arithmetic: the existence of the Lorenz attractor [38], the CAPD library for ordinary differential equations [39], KAM theory applied by computer [40], among other systems to magnetic field lines [41], blow-up of the Boussinesq and axisymmetric Euler equations [42], and elliptic and parabolic problems [43], among them the existence of stationary Navier–Stokes flows in three-dimensional domains near a computed approximation [44]. These proofs run in libraries outside a proof assistant. Inside one, the closest work is the verified ODE solvers of Isabelle/HOL up to the Lorenz attractor [45, 46]; the discretization and floating-point program of the one-dimensional wave equation [47], the Lax–Milgram theorem [48] and simplicial Lagrange finite elements [49] in Rocq, with floating-point error analysis in VCFLOAT2 and LAPROOF [50, 51]; and the certificates of the Kepler conjecture [52]. In Lean, high-energy physics is being formalized [53], a formalization has found an error in the literature on the stability of the two-Higgs-doublet potential [54], the characterization of the equilibria of the Vlasov–Maxwell–Landau system has been formalized [55], and smooth magnetohydrostatic equilibria with nested pressure surfaces and only a discrete rotational symmetry, counterexamples to Grad’s conjecture [17], come with a Lean certification [56]. In fusion, formal methods have verified a plasma vertical-position control algorithm [57] and properties of neural surrogates of the Grad–Shafranov equation [58]. Stellarator equilibrium codes are verified by convergence studies, manufactured solutions and comparison between codes: VMEC against SPEC [23, 24, 20] and against DESC [59, 16], and against Landreman’s explicit three-dimensional equilibria [60], whose supplement solves them in DESC. We know of no prior work that states the output of a stellarator equilibrium code as a machine-checked theorem about the field reconstructed from it.

7 Conclusion

Stellarocq gives the output of an equilibrium code a meaning a proof assistant can check: bounds, harmonics, Mercier stability terms and the existence of discrete solutions, each a theorem about the field reconstructed from the file’s exact numbers, whose residual at a free radius is proven to be the ideal-MHD force. Against Landreman’s explicit equilibria [60], both of whose families the development proves to be ideal-MHD equilibria, the same certificates measure a solver’s error against a true equilibrium and not a manufactured one.

Floating point gives most of these numbers to several digits; the certificates add statements of another kind. A verdict is a theorem that rounding cannot overturn: at 17 surfaces the Mercier criterion is proven negative for the reconstruction and positive for the file’s own differences. A check quantifies over sets that no finite run samples, every state in a box of coefficients and every vacuum pressure between the answers of two solvers. It proves what no run shows, that a discrete solution exists in a given box and that Landreman’s fields are exact equilibria, whose distance from a solver’s output is then a true error. Anyone can check a certificate again with the released checker, without trusting the program that wrote it.

Limits. The certified field is the reconstruction’s, and its Hermite extension between half points is Stellarocq’s choice. VMEC’s discrete field differs from it by the products of Section 2, whose size on these files is evaluated in floating point, and the manufactured problems test Stellarocq’s collocation of the half-grid rule, not VMEC’s variational equations. A certificate describes the file it reads, not a converged equilibrium. A small residual does not by itself place an equilibrium nearby,

as fields sustained by a small force show [61]. The free-boundary statements hold for every vacuum pressure between NESTOR’s and BIEST’s answers, not for an exact vacuum field, and refine the surfaces with NESTOR’s grid fixed. The Mercier criterion is certified on the tokamak and not on W7-X, where a covering of 64×16 cells leaves its enclosures unbounded. A check costs seconds at points and hours for a harmonic at 256 bits or a projection of W7-X at $m \leq 19$ (Table 1), and every verdict rests on the trust base of Section 3.

Availability. The development and the certificate generators are at <https://github.com/CharlesCNorton/stellarocq>, where `make all` builds the proofs and the checker and each generator’s documentation gives the commands of its result. Its release `v1.0` holds the checker as a static Linux executable and the certificates behind every certified value of Sections 3 and 4, with the VMEC++ files they were written from, the command that checks each and the verdict it received. The VMEC++ tests that carry the certified values are on the branch of [28].

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